

# Midterm Exam 2 (Version 2)

CAS CS 132: Geometric Algorithms

Name:

BUID:

- ▷ You will have approximately 75 minutes to complete this exam. Make sure to read every question, some are easier than others.
- ▷ Do not remove any pages from the exam.
- ▷ Your final solution must appear in the solution boxes for each problem. **Only include your final solution in the solution box. You must show your work outside of the solution box.** You will not receive credit if you don't show your work.
- ▷ We will not look at any work on the pages marked “*This page is intentionally left blank.*” You should use these pages for scratch work.

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# 1 True/False

(10 points) Determine if each of the following statements is **True** or **False**. Bubble in your answers below.

- A. (1 point) For matrices  $A$  and  $B$  in  $\mathbb{R}^{m \times n}$ , let  $\begin{bmatrix} A \\ B \end{bmatrix}$  denote the  $\mathbb{R}^{2m \times n}$  obtained by stacking  $A$  and  $B$  vertically.

For any vector  $\mathbf{b} \in \mathbb{R}^n$ , if  $A\mathbf{x} = \mathbf{b}$  is consistent and  $B\mathbf{x} = \mathbf{b}$  is consistent then so is  $\begin{bmatrix} A \\ B \end{bmatrix} \mathbf{x} = \begin{bmatrix} \mathbf{b} \\ \mathbf{b} \end{bmatrix}$ .

- ☐ True  
☐ False

- B. (1 point) For any stochastic matrix  $A \in \mathbb{R}^{n \times n}$ , if  $A$  is regular then  $\dim(\text{Nul}(A - I)) = 1$ .

- ☐ True  
☐ False

- C. (1 point) For a subspace  $H$  of  $\mathbb{R}^n$  and vector  $\mathbf{v} \in \mathbb{R}^n$ , if  $\mathbf{v} \notin H$ , then  $\dim H < n$ .

- ☐ True  
☐ False

- D. (1 point) If  $A$  is invertible, then both the  $L$ -part and the  $U$ -part of an  $LU$ -factorization of  $A$  are invertible.

- ☐ True  
☐ False

- E. (1 point) Given linear transformations  $S : \mathbb{R}^m \rightarrow \mathbb{R}^n$  and  $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ , if  $m < n$ , then it is possible for  $T \circ S$  to be invertible (recall that  $(T \circ S)(\mathbf{v}) = T(S(\mathbf{v}))$ ).

- ☐ True  
☐ False

- F. (1 point) If  $\{\mathbf{v}_1, \mathbf{v}_2\}$  and  $\{\mathbf{v}_1, \mathbf{v}_3\}$  and  $\{\mathbf{v}_2, \mathbf{v}_3\}$  are linearly independent, then so is  $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ .

- ☐ True  
☐ False

- G. (1 point) For any stochastic matrix  $A \in \mathbb{R}^{n \times n}$ , if  $A$  has a unique steady-state distribution, then  $A$  is regular.

- ☐ True  
☐ False

- H. (1 point) If  $\text{rank } A = 1$ , then the matrix equation  $A\mathbf{x} = \mathbf{0}$  has infinitely many solutions.

- ☐ True  
☐ False

- I. (1 point) For a matrix  $A \in \mathbb{R}^{2026 \times 2027}$ , it is possible for the rank of  $A$  to be the same as  $\dim(\text{Nul } A)$ .

- ☐ True  
☐ False

- J. (1 point) For matrices  $A, B, C$  in  $\mathbb{R}^{n \times n}$ , if  $AB = AC$  then  $B = C$ .

- ☐ True  
☐ False

## 2 Matrix Inverses

$$A = \begin{bmatrix} 1 & -2 & 3 \\ 1 & 0 & 2 \\ -2 & 5 & -7 \end{bmatrix}$$

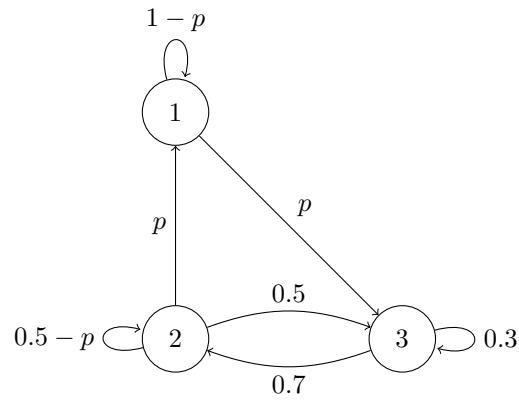
- A. (7 points) Determine the inverse of  $A$ , i.e., determine the matrix  $B$  such that  $AB = BA = I$ .

*Solution.*

- B. (3 points) Use the matrix you determined in Part A to calculate the solution to the matrix equation  $A\mathbf{x} = [1 \ 1 \ 1]^T$ . For partial credit, determine the solution to the matrix equation  $A\mathbf{x} = [1 \ 1 \ 1]^T$  using Gaussian elimination.

*Solution.*

### 3 Markov Chains



- A. (2 points) Determine the transition matrix  $A$  of the above state diagram in terms of  $p$ . The transitions out of state 1 should appear in the first column of  $A$ , out of 2 in the second column, and out of 3 in the third column.

*Solution.*

- B. (2 points) Is the matrix  $A$  regular in the case that  $p = 0$ ? Justify your answer.

*Solution.* ☐ Yes ☐ No

*Justification.*

C. (4 points) Determine  $\lim_{k \rightarrow \infty} A^k [0.2 \ 0.4 \ 0.4]^T$  in the case that  $p = 0.1$ .

*Solution.*

- D. (2 points) Determine  $\lim_{k \rightarrow \infty} A^k [0.2 \ 0.4 \ 0.4]^T$  in the case that  $p = 0$ . For partial credit, determine a general form solution that describes the steady-state vectors of  $A$  in the case that  $p = 0$  (the steady state vectors of a matrix  $A$  are the vectors  $\mathbf{v}$  such that  $A\mathbf{v} = \mathbf{v}$ ).

*Solution.*



## 4 Subspaces

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} \mapsto \begin{bmatrix} x + y - 3z \\ -x + 4z \\ 2x + y - 7z \end{bmatrix}$$

- A. (2 points) Determine the matrix  $A$  that implements the above transformation, i.e., determine the matrix  $A$  such that the above transformation is equivalent to  $\mathbf{x} \mapsto A\mathbf{x}$ .

*Solution.*

- B. (4 points) Determine a basis for  $\text{Nul } A$ .

*Solution.*

*You may also use this page to show your work for both Part B and Part C if necessary.*

C. (2 points) Determine a basis for  $\text{Col } A$ .

*Solution.*

D. (2 points) Let  $T$  denote the transformation from Part A and let  $H$  be the subspace

$$\{\mathbf{v} : \mathbf{v} \in \text{ran } T \text{ and } \mathbf{e}_3^T \mathbf{v} = 0\}$$

That is,  $H$  is **the subspace of vectors in the range of  $T$  whose third components are 0**. Note that  $H$  is a subspace of  $\text{ran } T$ , which is a subspace of  $\mathbb{R}^3$ . Determine a basis for  $H$ .

*Solution.*

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